

Financial Performance Analytics Platform

An end-to-end Business Intelligence case study — from raw business requirements to executive dashboard

Tech Stack	MySQL → Power Query → Power BI Desktop
Data Scope	3 fiscal years (FY2022–FY2024) · 5 departments · 360 fact rows
Deliverables	6 SQL analytical scripts · 1 ETL pipeline · 6-page interactive dashboard · 12+ DAX measures
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This project models three years of departmental P&L data for a fictional mid-sized retail company, builds a full SQL-to-dashboard analytics pipeline, and surfaces a \$15.6 Million cost overrun and a repeating seasonal budgeting error that had gone undetected across three fiscal years.

The goal was to demonstrate the complete lifecycle of a Business Analyst engagement: understanding a business problem, designing a data model to answer it, writing production-quality SQL, and delivering insights a CFO could act on.

1. Executive Summary

Meridian Retail Group is a fictional mid-sized retail organisation used as the basis for this project. Before this analysis, the company's financial data existed across three disconnected spreadsheets with no single source of truth, requiring three days of manual reconciliation at every month-end close.

This project builds a complete analytics pipeline — a MySQL star schema, six SQL scripts, a Power Query ETL layer, and a six-page Power BI dashboard — to give the CFO, Finance Manager, and five department heads a single, reliable view of revenue, budget performance, cost discipline, and departmental profitability.

Project At-a-Glance

Industry	Retail / Consumer Goods
Departments	Sales, Marketing, Operations, Human Resources, IT
Data Volume	360 fact rows · 36 months · 5 departments · 2 scenarios (Actual + Budget)
Dashboard	6 pages · 6 source tables · 12+ DAX measures · interactive slicers
Headline Finding	\$15.6 Million Marketing cost overrun · recurring seasonal budget miss across all 3 years

2. The Business Problem

The CFO's Core Pain Point

The project was scoped around a documented complaint from the CFO: revenue, cost, and budget data lived in three separate spreadsheets, each maintained by a different person, with no reliable way to know whether the company was on track, which department was overspending, or what the real profit margin was at any point in the year.

Specific Challenges

- **No single source of truth** — financial data spread across three disconnected files with no version control, producing conflicting totals and a 3-day manual month-end reconciliation.
- **No budget variance visibility** — departments had no real-time view of performance against budget; issues surfaced only at quarter-end, too late for corrective action.
- **No profitability ranking** — revenue and cost were tracked separately, so the CFO could not identify which departments created the most value.
- **No time-intelligence** — no mechanism existed to track year-to-date performance, month-over-month momentum, or rolling trends; all reporting was point-in-time.
- **Cost overrun blindness** — cost overruns were identified only at annual review, with no proactive in-year monitoring.

Stakeholder Impact

CFO	No consolidated financial view	Unable to make informed budget decisions
Finance Manager	3 days/month lost to manual reconciliation	Analytical time wasted on data prep instead of analysis
Sales Head	No visibility into budget tracking	Reactive rather than proactive management
Operations Head	No cost efficiency benchmarks	Cannot compare performance against plan
HR / IT Heads	No cost overrun alerts	Cost discipline issues detected too late

3. Data Architecture

Star Schema Design

The database uses a star schema — the standard model for financial analytics — with a central fact table connected to three dimension tables that provide filtering and grouping context.

fact_financials	Central transaction table	360 rows — 1 row per department × month × scenario
dim_date	Time dimension	36 rows — Year, Quarter, Month, Month Name
dim_department	Department dimension	5 rows — Dept name, head, region, cost-centre flag
dim_scenario	Scenario dimension	2 rows — Actual / Budget

Why the Scenario Dimension Matters

Storing Actual and Budget data in the same fact table via a **scenario_id** column is the most important design decision in the schema. It reduces all variance analysis to a simple CASE WHEN pivot — no separate tables, no complex joins, no risk of double-counting. This single choice enables Budget-vs-Actual comparison in one query, slicer-driven scenario selection in Power BI, and DAX measures that calculate variance dynamically based on filter context.

Summary Tables vs. Relational Model in Power BI

The Power BI layer consumes six pre-aggregated summary tables (the outputs of the SQL scripts) rather than the raw fact and dimension tables. This was a deliberate choice: each dashboard page loads only the data it needs, analysis on one page (e.g. revenue trends) can't be accidentally contaminated by another (e.g. cost analysis), and each page's slicers filter only their own table — eliminating the cross-table filtering conflicts that a fully relational model would introduce.

4. Analytical Approach — SQL Scripts

Six SQL scripts were developed against the MySQL database, each answering one specific business question. Outputs are exported as CSVs and consumed by Power BI via Power Query.

01 · Revenue Trends

Business question: How has company-wide revenue moved month-by-month across three fiscal years?

Technical approach: Excludes Budget rows to prevent double-counting; calculates Gross Profit and Net Margin % directly in SQL with NULLIF protection against division-by-zero.

Finding: Revenue grew from \$148M (FY2022) to \$155M (FY2024), roughly 2.3% CAGR. December is consistently the highest-revenue month every year.

02 · Budget Variance

Business question: Which departments hit their revenue and cost targets, which missed, and by how much?

Technical approach: Uses a CTE to pivot Actual and Budget into side-by-side columns via CASE WHEN, then computes variance in the outer query — a two-step approach chosen for readability and maintainability.

Finding: All five departments missed their revenue budgets in all three years; the shortfall grew from -\$731K (FY2022) to -\$1.98M (FY2024). December consistently beats budget while Jan–Mar consistently miss it.

03 · Department Profitability

Business question: Which departments are most profitable, and how do gross margin and cost efficiency rank across all five?

Technical approach: Uses RANK() OVER (PARTITION BY Year ORDER BY Gross Profit DESC) to rank departments within each year. An early version had an operator-precedence bug (Revenue – Cost * 100 / Revenue without parentheses) that produced nonsensical margins — corrected with explicit parentheses.

Finding: Sales ranks 1st (37% gross margin); HR (5%) and IT (8%) are confirmed cost centres requiring strict cost control rather than revenue expectations.

04 · Monthly KPIs (Window Functions)

Business question: Is revenue accelerating or decelerating? What are YTD performance and month-over-month growth?

Technical approach: Uses SUM() OVER and LAG() OVER, both partitioned by department, to calculate YTD revenue and MoM growth. An early version omitted PARTITION BY, causing one department's December to be compared against another department's November — fixed by partitioning and ordering explicitly.

Finding: Company-wide average MoM growth is ~2.3%, with September showing consistent deceleration and December the strongest growth spike every year.

05 · Cost Analysis

Business question: Which departments exceeded their cost budgets, by how much, and is the trend improving or worsening?

Technical approach: Pivots Actual and Budget cost via CASE WHEN, then computes Cost Overrun and Overrun % against the approved budget, with a categorical Budget Status flag for dashboard conditional formatting.

Finding: All five departments exceeded cost budgets in all three years. Marketing shows the highest absolute overrun (\$15.6M over 3 years); IT shows the highest percentage overrun (12.16% in FY2024).

06 · Executive Summary

Business question: One clean row per year, showing all headline financial KPIs for board-level reporting.

Technical approach: Pre-calculates annual totals into a 3-row table rather than aggregating a 360-row fact table on every dashboard render — a deliberate performance optimisation for the KPI cards.

Finding: Budget achievement declined from 98.2% (FY2022) to 97.3% (FY2024); net margin compressed from 29.2% to 28.7% over the same period.

5. Dashboard Overview

The dashboard comprises six pages, each built from a single SQL summary table — a one-table-per-page architecture that keeps slicers precise without relying on table relationships.

Page 1 • Revenue Overview — CEO, Board

Four KPI cards (Revenue, Cost, Gross Profit, Net Margin %), a 3-year monthly revenue line chart, a net margin area chart, and a quarterly gross profit bar chart. Confirms steady year-on-year revenue growth at the monthly level, with margin compression visible in the final months of FY2024.

Page 2 • Budget vs. Actual — CFO, Finance Manager

KPI cards for Actual/Budget/Variance %, a combo chart pairing absolute variance with variance rate, and a matrix table for board reporting. Shows the total 3-year shortfall of $-\$4.39\text{M}$ against a $\$459\text{M}$ budget, with the miss rate worsening every year.

Page 3 • Cost Intelligence — CFO, Operations Head

Cost KPI cards, a percentage-overflow bar chart, an absolute-overflow bar chart, and a share-of-total donut chart. Demonstrates that IT has the worst overflow by percentage ($\sim 12\%$) while Marketing causes the most damage in absolute terms ($\$15.6\text{M}$, $\sim 39\%$ of total overflow).

Page 4 • Revenue Intelligence — Finance Manager

Monthly variance % line chart, an Actual-vs-Budget dual line chart, and a variance detail matrix. Confirms a stable seasonal pattern: January is structurally the weakest month (-4.64% average variance), December structurally the strongest ($+2.34\%$).

Page 5 • Revenue Momentum — Sales Leadership, Strategy

MoM growth % bar chart with a zero reference line, a revenue-vs-rolling-average combo chart, and a YTD progression area chart. Confirms September as a recurring deceleration point and shows FY2024 tracking ahead of both prior years at every month.

Page 6 • Profitability Rank — Executive Team, Dept. Heads

Gross margin and cost-to-revenue bar charts plus a full profitability scorecard matrix. Ranks Sales 1st (37% margin, $\$0.63$ cost-to-revenue ratio) and confirms HR and IT as structural cost centres.

6. Key Insights & Recommendations

Insight 1 · Revenue growth is decelerating while costs grow faster

Revenue grew at ~2.3% CAGR over three years, but cost grew at ~4.1% CAGR — the direct cause of the net margin compression seen on Page 1.

Recommendation: Pair revenue-acceleration initiatives in Q1 (historically the weakest quarter) with company-wide cost controls.

Insight 2 · Seasonal budget misalignment repeats every year

January–March miss the revenue budget by 3–5% every year; October–December beat it by 1–2% every year — an identical pattern across all three fiscal years, pointing to a budgeting-process flaw rather than an execution failure.

Recommendation: Rebuild the annual budget with seasonal adjustments — reduce H1 targets by 3–5%, increase H2 targets by 1–2% — projected to lift budget achievement from ~98% to 99%+.

Insight 3 · Universal cost overruns indicate a systemic issue

All five departments exceeded their cost budgets in every year analysed — not an isolated department problem, but evidence that budget assumptions don't reflect operational reality.

Recommendation: Move from top-down budget allocation to zero-based budgeting, where each department builds its budget from planned activities rather than last year's figure plus inflation.

Insight 4 · Marketing requires an immediate ROI review

Marketing has the highest absolute cost overrun (\$15.6M over 3 years) while simultaneously missing its revenue targets — the only department underperforming on both fronts at once.

Recommendation: Run a campaign-level ROI analysis; any overspend not linked to measurable revenue lift represents pure value destruction and should be reallocated.

Business Value Delivered

- **Time saved:** eliminates ~36 days per year of Finance Manager time previously spent on manual reconciliation.
- **Decision speed:** budget variance visible within 24 hours of month-end close, vs. a prior 3-week delay.
- **Risk detection:** cost overruns identified at the department-month level in near real time, instead of only at annual review.
- **Strategic clarity:** a definitive profitability ranking to guide investment prioritisation across departments.
- **Audit trail:** every calculation is version-controlled in SQL, giving full analytical lineage from source data to executive insight.

7. Technical Architecture Summary

1	MySQL Database	Star schema — fact_financials + 3 dimension tables, 360 fact rows
2	SQL Scripts (×6)	Analytical summary tables exported as CSV; each answers one business question
3	Power Query	ETL pipeline — data type correction, column renaming, calculated columns, row validation
4	Power BI Model	6 independent summary tables, 12+ DAX measures, edit-interactions for slicer coordination
5	Dashboard	6 pages × 6 tables, distinct audiences per page, navigation and conditional formatting

DAX Measure Organisation

All DAX measures live in a dedicated `_Measures` table rather than being scattered across source tables, and follow a page-number naming prefix (e.g. `4.Revenue Variance`) to avoid naming conflicts between similarly-named measures on different pages.

Why Independent Tables Instead of Relationships

Six independent summary tables were used instead of a fully relational Power BI model for three reasons: **analytical isolation** (a revenue-page calculation can never be contaminated by a cost-page calculation), **slicer precision** (each page's filters apply only to its own table, avoiding cross-table filter propagation), and **performance** (pre-aggregated tables load and filter faster than computing aggregations from the 360-row fact table on every DAX evaluation).

Key Achievements

- Designed a MySQL star schema and wrote 6 SQL scripts using CTEs, CASE WHEN pivoting, and window functions (LAG, RANK, SUM OVER PARTITION BY) to produce analysis-ready summary tables.
- Built a Power Query ETL pipeline with explicit data-type handling, null handling, and row-count validation to ensure zero data-quality issues before dashboard load.
- Delivered a 6-page interactive Power BI dashboard with 12+ DAX measures, slicers, conditional formatting, and page navigation, serving 7 distinct stakeholder roles.
- Identified a \$15.6 Million Marketing cost overrun and a recurring seasonal budget-misalignment pattern, translated into 4 written executive recommendations with quantified business impact.

8. Financial Terms Glossary

Revenue	Total income earned from selling goods or services, per department.
Gross Profit	Revenue minus Total Cost — profit before tax and overhead adjustments.
Net Margin %	Gross Profit ÷ Revenue, expressed as a percentage — measures overall efficiency.
Budget / Actual	Budget: pre-approved targets set at the start of the fiscal year. Actual: real recorded figures for the period.
Budget Variance	Actual minus Budget — positive means the target was exceeded, negative means it was missed.
Variance %	Budget Variance ÷ Budget — the standard measure of plan attainment.
YTD Revenue	Year-to-Date — cumulative revenue from January 1 through the current month.
MoM Growth %	Month-over-Month — percentage change in revenue versus the prior month.
Cost-to-Revenue Ratio	Total Cost ÷ Total Revenue — lower values indicate greater efficiency.
Star Schema	A database model with a central fact table surrounded by dimension tables.
CTE	Common Table Expression — a named temporary result set used to simplify complex SQL.
Window Function	A SQL function (e.g. LAG, RANK, SUM OVER) that calculates across a set of rows without collapsing them.
DAX	Data Analysis Expressions — the Power BI formula language used to build measures.
ETL	Extract, Transform, Load — the data pipeline process from SQL through Power Query to Power BI.

Meridian Retail Group is a fictional company created for the purpose of this project. All figures, names, and data are illustrative and used to demonstrate an end-to-end Business Analyst / Business Intelligence workflow.